CIS 9440 · Spring 2026 · Final Project Report

NYC Illegal Parking

Squeeze Index

Cross-mart analysis of ACE bus-camera violations and 311 illegal-parking complaints across New York City, built on a Kimball star schema in BigQuery with dbt and visualized in Looker Studio.

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SECTION 1

Introduction

Illegal parking remains one of the most persistent friction points in New York City's curb economy: it slows down buses, blocks bike lanes, fills emergency hydrants, and shows up year after year in the top complaint categories on 311. This project examines that problem through two complementary lenses — public reporting (NYC 311 service requests) and automated enforcement (MTA Bus Automated Camera Enforcement, or "ACE"). By integrating the two into a single dimensional warehouse and analyzing them side-by-side across borough, time, and day-of-week, we set out to identify where curb conflict is most acute and where the public's perception of illegal parking matches — or diverges from — actual enforcement activity.

Our original proposal framed this as the "Squeeze Index" — a search for geographic bottlenecks where static curb rules conflict with high-volume transit flow. Over the course of Milestones 2 through 5 we narrowed the scope to a clean, defensible cross-mart comparison: **5.5+ years of NYC 311 illegal-parking complaints alongside 6+ years of MTA ACE bus-camera violations**, both modeled in a shared Kimball star schema and joined on conformed Date and Location dimensions.

Analytics questions

Building on the framing in our Milestone 1 proposal, the final analysis focused on five concrete cross-mart questions, each implemented as one of the queries powering the Looker Studio dashboard:

- **Which boroughs experience the highest illegal-parking activity?** — comparing ACE violations and 311 complaints side-by-side at the borough level.
- **How has illegal-parking activity changed over time?** — tracking annual totals for both data sources.
- **Are there monthly or seasonal patterns in illegal parking?** — looking at the joint monthly trend of ACE and 311 from October 2019 onward.
- **Which days of the week see the most illegal parking?** — comparing the weekday-vs-weekend mix of ACE enforcement against citizen reporting.
- **What types of illegal-parking violations are most common across boroughs?** — a borough-level breakdown of ACE's three primary violation categories (Bus Stop, Bus Lane, Double Parking).

Stakeholders

Four groups stand to use the kind of analysis we built:

- **NYC Department of Transportation (DOT)** — the agency with regulatory authority over curb space and the lead on initiatives like the 2025–2026 Smart Curbs pilot. Our cross-mart view tells DOT where existing static curb rules are most strained.
- **Metropolitan Transportation Authority (MTA)** — operates ACE and uses its data to identify bus-corridor obstructions. Pairing ACE with 311 helps the MTA see where citizen perception aligns with what the cameras are recording.
- **NYPD Traffic Enforcement District** — handles manual ticketing and tow authorization. Borough- and day-of-week patterns can help calibrate enforcement deployment.
- **NYC transit riders and the general public** — the ultimate beneficiaries of better curb management, and the source of every 311 complaint in this dataset.

What might change

We do not claim to recommend a specific policy. We do claim that the dashboard makes a few patterns very visible — most notably an enormous jump in ACE activity in 2024–2025 driven by program expansion, a sharp weekday-vs-weekend split in ACE that is essentially absent from 311, and very different violation-type mixes across boroughs. Each of those would be a reasonable starting point for a deeper neighborhood- or corridor-level study, and several point toward where Dynamic Curb Management pilots might be tested first.

Data sources

Source	Description and URL
NYC 311 Service Requests	Citizen-reported service complaints, filtered to illegal-parking-related complaint types (double parking, blocked hydrants, blocked driveways, blocked bus stops). Each row is one individual complaint. data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9
MTA Bus ACE Violations	Bus-mounted camera detections of vehicles obstructing bus lanes, bus stops, or double-parked, beginning October 2019. Each row is one enforcement event. data.ny.gov/Transportation/MTA-Bus-Automated-Camera-Enforcement-Violations-Beginning-October-2019/kh8p-hcbm

KPIs reported on the dashboard

The dashboard reports two overall-scale scorecards plus the five cross-mart visualizations above:

- **Total ACE Violations** — single-number scorecard, full data window.
- **Total 311 Complaints (illegal parking)** — single-number scorecard, full data window.
- **ACE vs 311 by borough** — grouped column chart.

- **ACE vs 311 by year** — grouped column chart.
- **Monthly trend of ACE vs 311** — time-series line chart, the dashboard's primary cross-mart view.
- **ACE vs 311 by day of week** — grouped column chart.
- **ACE violation type mix across boroughs** — 100% stacked column chart (ACE-only).

A note on KPI evolution: our Milestone 1 proposal sketched four normalized KPIs (Curb Conflict Intensity, Enforcement Efficiency Ratio, Blocked Bus Stop Complaint Intensity, ACE Complaint Share) that required bus-stop geocoding to compute. When we examined the actual schemas during Milestone 2, the per-bus-stop join turned out to be unreliable — ACE bus-stop IDs and the public bus-stop registry did not align cleanly enough to produce defensible per-stop normalization. We pivoted to the five count-based cross-mart breakdowns above, which answer the same underlying "where, when, and what kind?" questions without depending on a fragile join.

SECTION 2

Dimensional Model

The final integrated dimensional model is a two-fact star schema in BigQuery (project **riya-gcp-1**, dataset **nyc_illegal_parking_marts**). Both fact tables share two conformed dimensions (Date and Location), and each has its own process-specific dimensions. The model is the actual structure we built in Milestone 4 with dbt — not a draft. Surrogate keys are used everywhere a natural key would be unstable across loads.

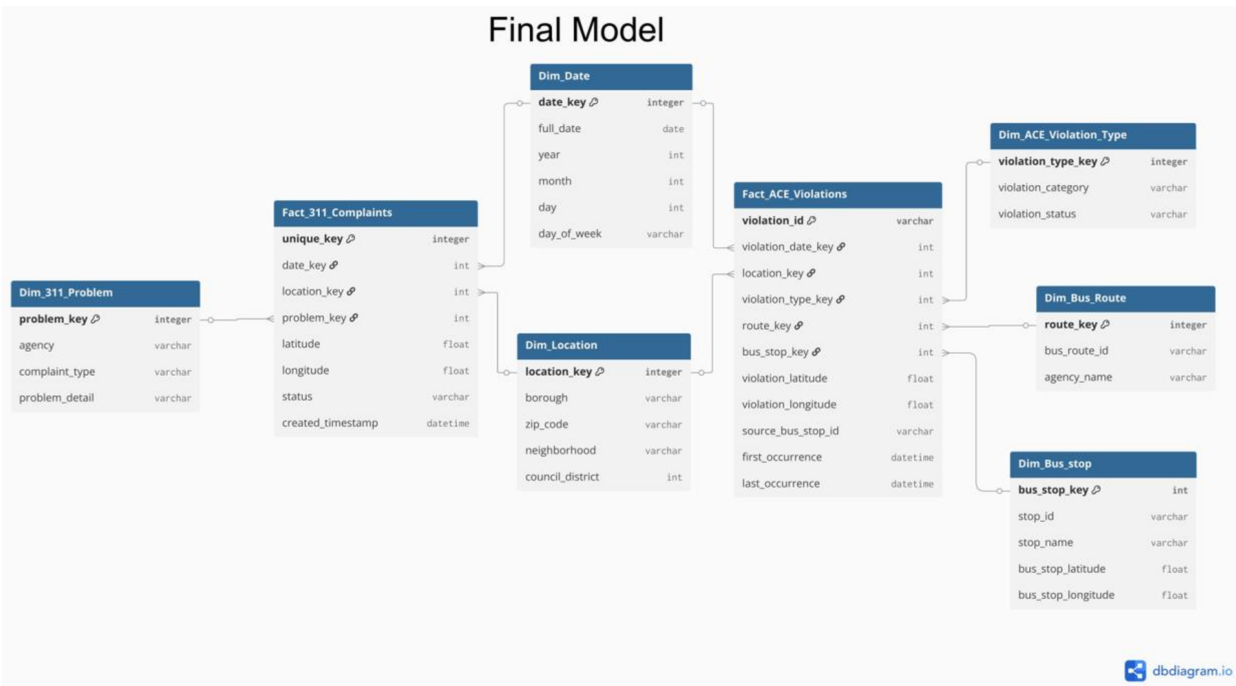

Figure 1. Final integrated dimensional model. Two fact tables share conformed Date and Location dimensions. 311-specific dimensions are on the left; ACE-specific dimensions are on the right.

Fact table grain

Fact table	Grain (one row =)
Fact_311_Complaints	One illegal-parking-related 311 service request (~2.55M rows).
Fact_ACE_Violations	One ACE camera-detected violation event (~6.36M rows).

Conformed dimensions (the cross-mart bridge)

Dim_Date and **Dim_Location** are the conformed dimensions that make every cross-mart query in this project possible. Both fact tables join to them, which lets us slice ACE and 311 by the same time and place attributes and compare them on equal footing.

- **Dim_Date** — integer surrogate key `date_key`; attributes include `full_date`, `year`, `month`, `day`, `day_of_week`. Granularity is one row per calendar day.
- **Dim_Location** — integer surrogate key `location_key`; attributes include `borough`, `zip_code`, `neighborhood`, `council_district`. Joined to both facts via `UPPER(TRIM(borough))` to normalize text differences between sources.

Process-specific dimensions

- **Dim_311_Problem** — distinct `(agency, complaint_type, descriptor)` tuples from the 311 source. Lets us slice 311 by problem detail.
- **Dim_ACE_Violation_Type** — distinct `violation_type` values from ACE; a CASE expression buckets each into one of three categories (Bus Stop, Bus Lane, Double Parking) plus an "Other" fallback.
- **Dim_Bus_Route** — distinct `bus_route_id` values from ACE (e.g., Bx19, M15-SBS).
- **Dim_Bus_Stop** — distinct `(stop_id, stop_name, bus_stop_latitude, bus_stop_longitude)` from ACE.

Modeling decisions and challenges

A few notes on the choices reflected in the diagram:

- **Two facts, not one consolidated fact.** ACE and 311 are different business processes with different grains and very different attribute sets. Forcing them into a single fact would have required either lossy harmonization or sparse columns. The two-fact pattern keeps each clean and uses the conformed dimensions to bridge them at query time.
- **Location was harder than it looked.** The two sources use different spellings, casing, and (in some 311 rows) NULL boroughs. We chose to standardize on borough as the join grain in the marts layer — finer-grained joins on zip or neighborhood were noisier and would have dropped rows. ZIP code and neighborhood are preserved in `Dim_Location` for future drill-downs but the marts dashboard uses borough.
- **Bus_stop did not become a third conformed dimension.** Our original M1 KPIs assumed we could normalize complaints and violations by bus stop. The ACE `bus_stop_id` field did not match cleanly against the public bus-stop registry, so we kept `Dim_Bus_Stop` as an ACE-only dimension and dropped the bus-stop-normalized KPIs. The two facts still share Date and Location, which is enough for the cross-mart views on the dashboard.

- **Degenerate dimensions and measures.** Both fact tables retain raw latitude/longitude as degenerate dimensions for future geospatial work. `Fact_311_Complaints` carries `status` and `created_timestamp` for SLA-style analysis; `Fact_ACE_Violations` carries `first_occurrence` and `last_occurrence` timestamps and the source `bus_stop_id` string.

SECTION 3

Data Processing & Technical Architecture

The project follows a standard ELT pattern — **Extract** → **Load** → **Transform** — with raw data landing in BigQuery first and dbt handling all transformations in versioned SQL. The result is a three-layer architecture (raw → staging → marts) that an analyst can rebuild end-to-end with a single `dbt build --select project_work` command.

Tools and what each was used for

Stage	Tool	Purpose
Extract	NYC / NY Open Data Socrata APIs (CSV)	Pull raw CSV files for the two source datasets directly from NYC Open Data and NY State Open Data.
Load	BigQuery web UI / bq CLI	Upload CSVs into the `nyc_proj2_raw_data` dataset, one table per source, mirroring the source schema 1:1.
Transform	dbt-bigquery + dbt-utils	Versioned SQL models that build staging views and mart tables. Models are defined in a public GitHub repo and reference each other through `ref()` so dbt builds the correct DAG.
Orchestrate	dbt build (manual)	Single command to run all models and tests in dependency order.
Warehouse	Google BigQuery	Compute + storage for raw, staging, and marts. Project: **riya-gcp-1**.
BI	Looker Studio	Connects directly to BigQuery via custom-SQL data sources, one per chart.

Three-layer ELT

Every dataset moves through the same three layers, with each layer materialized in its own BigQuery dataset:

Layer	BigQuery dataset	What lives here
Raw	`nyc_proj2_raw_data`	`source_nyc_311_traffic`, `source_mta_ace_violations`. CSVs loaded as-is. No transformations applied.
Staging	`nyc_illegal_parking_staging`	`stg_nyc_311_illegal_traffic`, `stg_mta_ace_violations`. Materialized as views. Casts types, trims/normalizes strings, filters 311 to illegal-parking-related complaint types and the last 7 years.

Layer	BigQuery dataset	What lives here
Marts	`nyc_illegal_parking_marts`	Eight tables: `dim_date_project`, `dim_location_project`, `dim_311_problem`, `dim_ace_violation_type`, `dim_bus_route`, `dim_bus_stop`, `fact_311_complaints`, `fact_ace_violations`. Materialized as tables.

Transform stage — dbt models

The dbt project organizes models into staging and marts subdirectories. Marts models are built in dependency order:

- **Conformed dimensions** — `dim_date_project` and `dim_location_project` are built first from the union of staged 311 and ACE dates and locations.
- **Process dimensions** — `dim_311_problem` (from 311 staging); `dim_ace_violation_type`, `dim_bus_route`, `dim_bus_stop` (from ACE staging).
- **Facts** — `fact_311_complaints` joins 311 staging to its dimensions; `fact_ace_violations` joins ACE staging to all five of its dimensions. Joins use `ref()` so dbt enforces correct build order.

Both facts join to the conformed dimensions using the same join keys (`full_date` for date; uppercased/trimmed `borough` for location), which is what guarantees the cross-mart counts in the dashboard are comparable.

Analytics layer — Looker Studio

Each chart in the dashboard is backed by its own BigQuery custom-SQL data source. The five analytics queries are all in the project's SQL file and follow a consistent UNION ALL pattern that returns three columns: `dimension`, `series` ('ACE Violations' or '311 Complaints'), and `event_count`. The two scorecards are separate single-row `COUNT(*)` queries. Keeping one data source per chart isolates each visualization, makes Looker's caching predictable, and keeps BigQuery cost predictable.

How to reproduce this work

To rebuild the full warehouse end-to-end, a colleague would:

- Clone the public GitHub repo at github.com/rabulhasan/CIS9440, branch `beginning-staging-work`.
- Set up a `profiles.yml` pointing at a BigQuery project with the three target datasets created (`nyc_proj2_raw_data`, `nyc_illegal_parking_staging`, `nyc_illegal_parking_marts`).

- Load the two source CSVs into `nyc_proj2_raw_data` (links to the source endpoints are in Section 1 and in the References).
- Run `dbt build --select project_work` from the project root. dbt will materialize the staging views and then the marts tables in dependency order, running schema and uniqueness tests along the way.
- Connect Looker Studio to the project via the BigQuery connector, recreating one custom-SQL data source per chart using the queries in the SQL file.

SECTION 4

Analysis & Visualization Summary

The dashboard is hosted on Looker Studio and is publicly viewable at the link below. It pairs two overall-scale KPI scorecards with five cross-mart visualizations.

Live dashboard: datastudio.google.com/s/ofQg6Rtwmfo

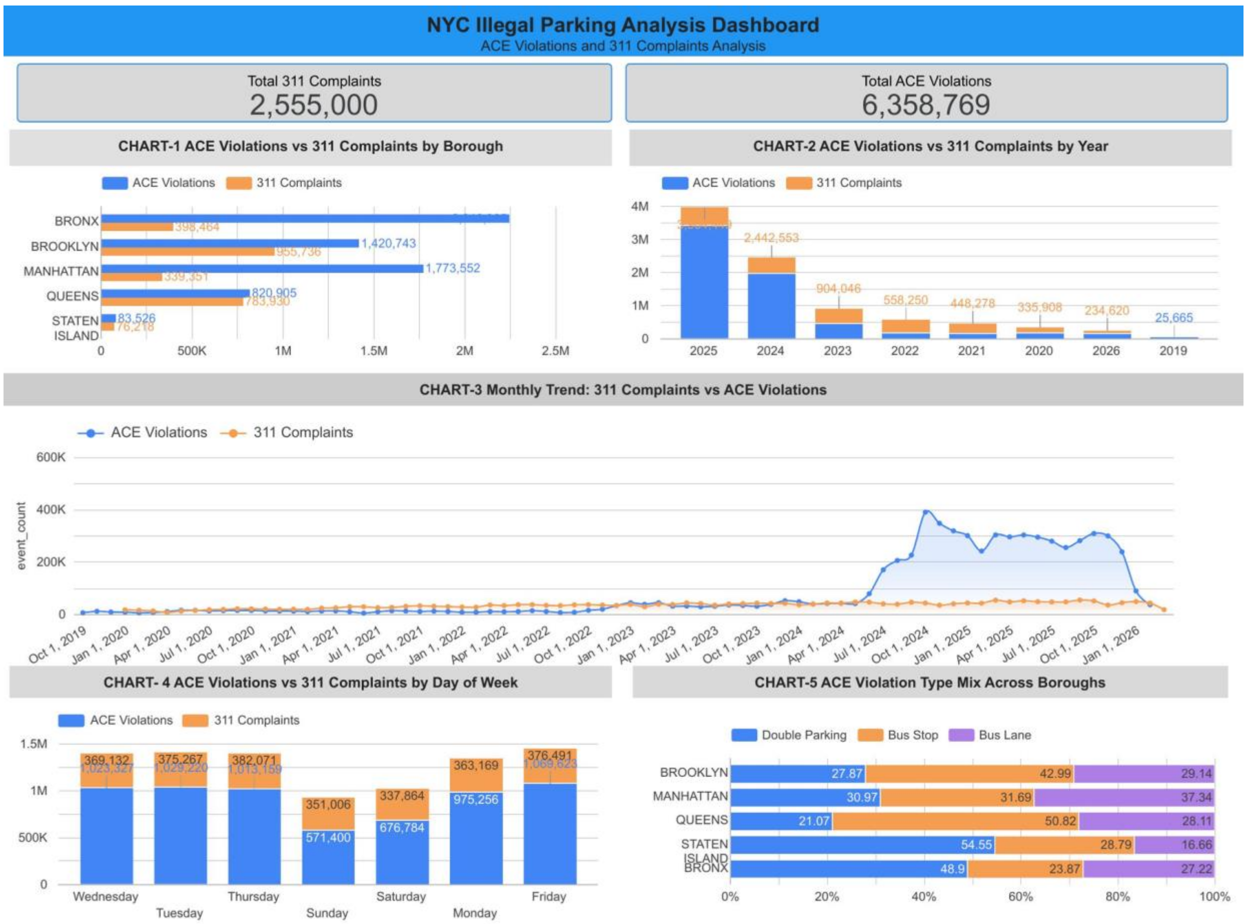

Figure 2. Full Looker Studio dashboard — two KPI scorecards and five cross-mart visualizations.

What each visualization shows

Scorecards: overall scale

- **Total ACE Violations: 6,358,769** — total bus-camera detections across the full window (Oct 2019–early 2026).
- **Total 311 Complaints: 2,555,000** — total illegal-parking-related citizen complaints. ACE outnumbers 311 by ~2.5×, which is consistent with automated cameras catching far more events than citizens report.

Chart 1 — ACE vs 311 by borough (grouped column)
Borough rankings differ sharply between the two sources. **Bronx leads ACE with 2.25M violations** (far ahead of Manhattan's 1.77M and Brooklyn's 1.42M) but trails on 311 (398K), reflecting that ACE camera coverage skews toward Bronx bus corridors. **Brooklyn leads 311 with 956K complaints.** Staten Island is small on both (84K ACE, 76K 311) and is the only borough where ACE and 311 are close in magnitude.

Chart 2 — ACE vs 311 by year (grouped column)
ACE volume has grown dramatically as the program has expanded: **904K (2023) → 2.44M (2024) → 3.96M (2025)** — a roughly four-fold jump in two years, almost certainly driven by camera deployment rather than a real change in driver behavior. 311 complaint volume across the same window has been comparatively flat. 2026 has only partial data (the dataset cut off in early 2026) so the apparent dip is incomplete coverage, not a real decline.

Chart 3 — Monthly trend: 311 vs ACE (time-series line)
The dashboard's primary cross-mart view. ACE is essentially flat near zero from Oct 2019 through mid-2023, then ramps sharply, peaking near **400K violations in a single month in late 2024**, and oscillating between 250K–350K through 2025. 311 stays flat around 30–50K per month across the entire window. The visual makes the ACE expansion story unmistakable.

Chart 4 — ACE vs 311 by day of week (grouped column)
ACE is heavily weekday-concentrated: **~1.0M violations on each weekday (Mon–Fri)**, dropping to 677K Saturday and 571K Sunday. 311 is nearly flat across all seven days (~350K–380K per day). The weekday-vs-weekend split in ACE is consistent with bus service patterns; the flatness in 311 is consistent with citizen reporting behavior being driven more by residential proximity than by transit schedules.

Chart 5 — ACE violation type mix across boroughs (100% stacked column)
This is the one ACE-only chart (311 has no comparable typology). The mix varies meaningfully by borough: **Staten Island is 55% double parking**; **Brooklyn is 43% Bus Stop**; **Manhattan is the most Bus-Lane-heavy at 37%**; **Bronx is 49% double parking with very little Bus Lane (23%)**. The mix likely reflects each borough's road geometry (Manhattan has the densest dedicated bus-lane network; outer-borough streets see more curb-blocking double parking).

SECTION 5

Narrative Conclusion & Results

Pulling the five visualizations together, three findings stand out — each pointing back to one of our original analytics questions, and each suggesting where additional analysis would be most valuable.

What the dashboard tells us

ACE growth is a program-expansion story, not a driver-behavior story. The four-fold jump in ACE violations between 2023 and 2025 dominates almost every other pattern in the data. 311 complaints over the same window are essentially flat. This means most of the cross-mart trend visible in the dashboard is being driven by where and when MTA chose to deploy more bus cameras — not by where illegal parking actually got worse. Any future analysis that compares pre-2024 and post-2024 data needs to control for this expansion or it will see "growth" everywhere.

Borough rankings flip depending on which source you ask. Bronx leads on ACE but trails on 311; Brooklyn leads on 311 but trails on ACE. The original Squeeze Index hypothesis was that high-conflict locations would show high values on both sources. What we see instead is that the two sources measure different things — ACE measures where buses encounter obstructions, which depends on camera coverage; 311 measures where residents are motivated enough to call. Staten Island is the only borough where the two are close, which is itself an interesting signal worth a closer look.

ACE and 311 have very different weekday-vs-weekend signatures. ACE drops by roughly half on weekends; 311 barely moves. This is a clean diagnostic for the kind of curb conflict each source detects: ACE is a transit-operations signal, 311 is a residential-quality-of-life signal. Combining them is more useful for triangulating problem locations than for averaging into a single index.

What stakeholders might do

We don't make specific policy recommendations, but the dashboard surfaces a few starting points:

- **For NYC DOT and MTA jointly:** the Bronx and Manhattan stand out as the boroughs where ACE finds the most violations but the violation-type mix is very different (Manhattan is bus-lane-heavy, Bronx is double-parking-heavy). Smart Curbs or Dynamic Curb Management pilots in those two boroughs would face very different design constraints and are worth analyzing as separate problems.

- **For MTA bus operations:** the weekday concentration of ACE plus the late-2024 peak in Chart 3 suggest specific corridors and time-of-day windows where buses are encountering the most curb obstruction. Drilling into Q3 monthly data at the bus-route level (using `Dim_Bus_Route`, which we built but didn't expose on the dashboard) is the obvious next step.
- **For NYPD Traffic Enforcement:** Brooklyn's lead in 311 volume without a matching ACE volume hints at "dark spots" where citizens are reporting illegal parking but ACE coverage is thin. A Brooklyn-focused complaint-density-by-zip analysis would help triage where manual enforcement could fill the gap.
- **For the general public and transit riders:** the dashboard is publicly viewable, so anyone wanting to see borough- or day-level patterns in their own neighborhood can.

Limitations and follow-up analysis we'd recommend

- **The two sources cover different time windows.** 311 goes back to 2010; ACE starts in October 2019. We trimmed 311 to the last 7 years to keep the two roughly comparable, but they still differ at the edges.
- **The borough-level join hides finer-grained patterns.** We joined on borough because zip- and neighborhood-level joins were noisier. A targeted neighborhood-level analysis (using the `zip_code` and `neighborhood` attributes already in `Dim_Location`) would be the natural follow-up.
- **Bus-stop normalization didn't work cleanly.** Our original M1 KPIs depended on normalizing complaints and violations by the number of bus stops. The ACE `bus_stop_id` field didn't match the public bus-stop registry reliably enough to do that. A future analysis that resolves bus-stop identity (probably via spatial join on lat/lon) could revive the Curb Conflict Intensity and Blocked Bus Stop Complaint Intensity KPIs from the proposal.
- **No geospatial visualization.** Both fact tables carry lat/lon as degenerate dimensions but the marts-level dashboard doesn't map them. A choropleth at the council-district or community-board level would make the borough-level patterns much more actionable.
- **ACE expansion confounds time trends.** As noted above, any year-over-year comparison needs to account for camera deployment. A useful follow-up would be to normalize ACE counts by the number of active ACE-equipped routes per month.

SECTION 6

References

Data sources

Both data sources are public datasets accessible without authentication. Accessed May 2026.

NYC 311 Service Requests (2010 to Present) — curator: NYC Open Data / NYC Department of Information Technology & Telecommunications. data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9

MTA Bus Automated Camera Enforcement Violations (Beginning October 2019) — curator: NY State Open Data / Metropolitan Transportation Authority. data.ny.gov/Transportation/MTA-Bus-Automated-Camera-Enforcement-Violations-Be/kh8p-hcbm

Warehouse and analytics tables

All warehouse tables live in BigQuery project **riya-gcp-1**. Professor Cohen (jaclyn.cohen@baruch.cuny.edu) has been granted BigQuery Data Viewer access to verify all tables. Access can be revoked after grading.

- **Raw layer:** `riya-gcp-1.nyc_proj2_raw_data` (`source_nyc_311_traffic`, `source_mta_ace_violations`).
- **Staging layer:** `riya-gcp-1.nyc_illegal_parking_staging` (`stg_nyc_311_illegal_traffic`, `stg_mta_ace_violations`).
- **Marts layer:** `riya-gcp-1.nyc_illegal_parking_marts` (`dim_date_project`, `dim_location_project`, `dim_311_problem`, `dim_ace_violation_type`, `dim_bus_route`, `dim_bus_stop`, `fact_311_complaints`, `fact_ace_violations`).

Code and dashboard

GitHub repository (dbt project, branch beginning-staging-work): github.com/rabulhasan/CIS9440

Looker Studio dashboard (publicly viewable; also shared with the professor): datastudio.google.com/s/ofQg6Rtwmfo

The five analytics SQL queries plus two scorecard queries are in the accompanying file `CIS_9440UTA_group_2_milestone5_queries.sql` (included in the appendix).

Software and platforms used

- **Google BigQuery** — cloud data warehouse (compute + storage).
- **dbt-bigquery** (with dbt-utils) — SQL-based transformation framework.

- **Looker Studio** (formerly Google Data Studio) — BI visualization layer.
- **dbdiagram.io** — used to draft the dimensional model diagram.
- **GitHub** — version control for dbt project.

AI tool use

Per academic-honesty policy, we disclose AI tool use here. AI-assisted work on this final report and the prior milestone documentation was limited to documentation drafting, PDF layout and formatting, and copy editing. The analytics queries, KPI definitions, dimensional model design, dbt model code, and Looker Studio dashboard were produced by the team. Specifically, an AI agent (Perplexity Computer) was used to draft the Milestone 4 ELT documentation, the Milestone 5 PDFs (intro, queries-work, dashboard guide), and this final report, based on the team's queries, KPIs, and dashboard. All AI output was reviewed, edited, and verified by the team before submission.